

Date: / / 20

# Assignment

Govt. Post Graduate College  
for women

BS Islamiyat B3-S2

Subject:

Quantitative Reasoning I  
(GQR-101)

Submitted by:

Group # 6

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## Factorial notation:

"Factorial notation is a mathematical notation used to represent the product of all positive integers from 1 up to a given number."

It denoted by ! mark

$$n! = n \times (n-1) \times (n-2) \times \dots \times 3 \times 2 \times 1.$$

$\therefore$  Exercise :-

$$\underline{\underline{2}} \quad 6!$$

$\underline{\underline{1}}$

$$4!$$

$$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 =$$

$$4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24$$

$\underline{\underline{3}}$

$$\frac{8!}{7!}$$

$$\underline{\underline{4}} \quad \frac{10!}{7!}$$

$$= \frac{8 \cdot 7!}{7!} = 8$$

$$= \frac{10 \cdot 9 \cdot 8 \cdot 7!}{7!} = 720$$

$\underline{\underline{5}}$

$$\frac{11!}{4! \times 7!}$$

$$= \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{4 \cdot 3 \cdot 2 \cdot 1 \cdot 7!} = 11 \cdot 10 \cdot 3 = 330$$



68!4! × 2!

$$= 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$4! \cdot 2 \cdot 1$$

$$= 8 \cdot 7 \cdot 3 \cdot 5 = 840$$

13!0!

$$= 3 \cdot 2 \cdot 1 = 6$$

1

811!2! 4! 5!

$$= 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$2 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 5!$$

$$11 \cdot 10 \cdot 9 \cdot 7 = 6930$$

99!2! (9-2)!

$$= 9 \cdot 8 \cdot 7!$$

2! (7)!

$$= 9 \cdot 8 \cdot 7!$$

2! 1 7!

$$9 \cdot 4 = 36$$

1015!15! (15-15)!

$$= 15!$$

15! (0)!

$$= \frac{1}{1 \cdot 1} = 1$$

116!3! 3!

$$= 6 \cdot 5 \cdot 4 \cdot 3!$$

3! 2! 1! 3!

$$= 2 \cdot 5 \cdot 2 = 20$$

124! 0! 1!

$$4 \cdot 3 \cdot 2 \cdot 1 \cdot 1 \cdot 1 = 24$$

each following factorial form

$$1 \quad \frac{6 \cdot 5 \cdot 4}{6!}$$

$$2 \quad \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1}$$

$$3 \quad \frac{12 \cdot 11 \cdot 10}{12!}$$

$$\frac{8!}{3!}$$

$$5 \quad \frac{20 \cdot 19 \cdot 18 \cdot 17}{20!}$$

$$4 \quad \frac{52 \cdot 51 \cdot 50 \cdot 49}{4 \cdot 3 \cdot 2 \cdot 1}$$

$$6 \quad \frac{10 \cdot 9}{2 \cdot 1}$$

$$\frac{52!}{4!}$$

$$\frac{10!}{2!}$$

$$8 \quad \frac{n(n-1)(n-2)}{n!}$$

$$7 \quad \frac{(n+2)(n+1)(n)}{(n+1)! (n-1)!}$$

Permutation:

"The arrangement of finite number of object one or more at a time is called permutation."

$${}^n P_r = \frac{n!}{(n-r)!}$$



## Exercise:-

$${}^n P_r = \frac{n!}{(n-r)!}$$

1.  ${}^{20} P_3$

$${}^{20} P_3 = \frac{20!}{(20-3)!} = \frac{20!}{17!} = \frac{20 \cdot 19 \cdot 18 \cdot 17!}{17!}$$

$$= 20 \cdot 19 \cdot 18 = 6840$$

2.  ${}^{16} P_4$

$${}^{16} P_4 = \frac{16!}{(16-4)!} = \frac{16!}{12!} = \frac{16 \cdot 15 \cdot 14 \cdot 13 \cdot 12!}{12!}$$

$$= 16 \cdot 15 \cdot 14 \cdot 13 = 43680$$

3.  ${}^{12} P_5$

$${}^{12} P_5 = \frac{12!}{(12-5)!} = \frac{12!}{7!} = \frac{12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7!}{7!}$$

$$= 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$$

4.  ${}^{10} P_7$

$${}^{10} P_7 = \frac{10!}{(10-7)!} = \frac{10!}{3!} = \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3!}{3!}$$

$$= 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 = 604800$$

5.  ${}^9 P_8$

$${}^9 P_8 = \frac{9!}{(9-8)!} = \frac{9!}{1!} = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 362880$$

Find the value  $n$ 

$$1. \quad {}^{11}P_n = 11 \cdot 10 \cdot 9$$

$$\frac{11!}{(11-n)!} = 11 \cdot 10 \cdot 9$$

$$(11-n)!$$

$$= \frac{11!}{(11-n)!} = 990 \Rightarrow (11-n)! = \frac{11!}{990}$$

$$= (11-n)! = 8!$$

$$11-n = 8 \Rightarrow 11-8 = n=3$$

$$2. \quad {}^nP_4 : {}^{n-1}P_3 = 9:1$$

$$\frac{{}^nP_4}{{}^{n-1}P_3} = 9 \Rightarrow \frac{\frac{n!}{(n-4)!}}{\frac{(n-1)!}{(n-1-3)!}} = 9$$

$$\frac{n!}{(n-4)!} \cdot \frac{(n-4)!}{(n-1)!} = 9$$

$$= \frac{n!}{(n-1)!} = 9 \Rightarrow \frac{n(n-1)!}{(n-1)!} = 9$$

$$n=9$$

\* How many signals can be given by 5 flags of different colours, using 3 flags at a time?

Total number of flags = 5

Number of signals using = 3 flags  
 $= {}^5P_3$

\* How many signals can be given by 6 flags of different colours when any number of them are used at a time

Total number of flags = 6

$$\frac{6!}{n!} = \frac{6 \cdot 5 \cdot 4!}{4!} = 6 \cdot 5 = 30$$

Number of signal using one flag =  ${}^6P_1$   
 $= \frac{6!}{(6-1)!} = \frac{6 \cdot 5!}{5!} = 6$

Number of signal 2 flags =  ${}^6P_2$   
 $= \frac{6!}{(6-2)!} = \frac{6 \cdot 5 \cdot 4!}{4!} = 6 \cdot 5 = 30$

Number of signal 3 flags =  ${}^6P_3$   
 $= \frac{6!}{(6-3)!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3!} = 6 \cdot 5 \cdot 4 = 120$



Number of signal 4 flags =  ${}^6P_4$

$$\frac{6!}{(6-4)!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2!} = \frac{6 \cdot 5 \cdot 4 \cdot 3}{1} = 360$$

Number of signal 5 flags =  ${}^6P_5$

$$\frac{6!}{(6-5)!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1} = 720$$

Number of signal 6 flags =  ${}^6P_6$

$$\frac{6!}{(6-6)!} = \frac{6!}{0!} = \frac{6!}{1} = 6! = 720 \quad (\because 0! = 1)$$

$$6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$$

Total number of signals

$$6 + 30 + 120 + 360 + 720 + 720 = 1956$$

\* How many words can be formed from the letter of the following words using all letter when no letter is to be repeated

1. PLANE 2. OBJECT 3. FASTING

→ PLANE  ${}^5P_5$

$$\frac{5!}{(5-5)!} = \frac{5!}{0!} = \frac{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1} = 120$$

→ OBJECT

$$\frac{6!}{(6-6)!} = \frac{6!}{0!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{1} = 720$$



⇒ FASTING

$${}^7P_7 = \frac{7!}{(7-7)!} = \frac{7!}{0!} = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$$

\* How many 3-digit numbers can be formed by using each one of the digits 2, 3, 5, 7, 9 only once?

Total number of digits = 5

$${}^5P_3 = \frac{5!}{(5-3)!} = \frac{5!}{2!} = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \div 2 \cdot 1 = 5 \cdot 4 \cdot 3 = 60$$

\* The Governors of the Punjab call a meeting of 12 officers. In how many ways can they be seated at a round table?

12 officers can be seated around the round table in  $11!$  ways

$$11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 39916800 \text{ ways}$$

\* The D.C.Os of 10 districts meet to discuss the law and order situation in their districts. In how many ways can they be seated at a round table when two particular D.C.Os insist on sitting together?

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 $= 987 \cdot 654 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 = 165100$   
 \* 11 members of a cube form 4 committees of 3, 4, 3, 2, 2 members so that one committed. Find the number of committees.

Total members = 11  
 3, 4, 2, 2 members of 4 committees and not repeated.

$$= 11!$$

$$\frac{3! \cdot 4! \cdot 2! \cdot 2!}{5!} = \frac{6 \cdot 24 \cdot 2 \cdot 2}{120} = \frac{480}{120} = 4$$

$$= 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$8 \cdot 7 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \cdot 2 \cdot 1$$

$$11 \cdot 5 \cdot 3 \cdot 4 \cdot 3 \cdot 7 \cdot 5 = 69300$$

Find the numbers of ways in which 5 men and 5 women can be seated at a round table in such a way that no two persons of the same sex sit together.

Let us first seat the men at a round table. We fix the seat of one man the remaining 4 of them can be seated in  $4!$  ways. Now we can seat 5 women one between two of the men in  $5!$  ways.



Thus 5 men and 5 women can be alternately seated at a round table in  $4! \times 5!$

$$= 4 \cdot 3 \cdot 2 \cdot 1 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 2880 \text{ ways}$$

Fatima invites 14 people to a dinner. There are 9 males and 5 females who are seated at two different tables so that guests of one sex sit at one round table and the guest of the other sex at the second table. Find the number of ways in which all guests are seated.

9 males can be seated at a round table in  $8!$  ways

5 females can be seated at a round table in  $4!$  ways

Both males and females can be seated in  $8! \times 4!$  ways.

$$= 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 967680 \text{ ways}$$

# Combination

An arrangement of  $n$ -different objects taken " $r$ " at a time without any order is called combination.

$${}^nC_r = \frac{n!}{(n-r)! r!}$$

∴ Exercise:-

$${}^{12}C_3$$

$$\frac{12!}{3!(12-3)!} = \frac{12 \cdot 11 \cdot 10 \cdot 9!}{3 \cdot 2 \cdot 1 \cdot 9!}$$

$$4 \cdot 11 \cdot 5 = 220$$

$${}^{20}C_{17}$$

$$\frac{20!}{17(20-17)!} = \frac{20 \cdot 19 \cdot 18 \cdot 17!}{17! \cdot 3 \cdot 2 \cdot 1}$$

$$10 \cdot 19 \cdot 6 = 1140$$

$${}^nC_4$$

$$\frac{n!}{4!(n-4)!} = \frac{n(n-1)(n-2)(n-3)(n-4)!}{4 \cdot 3 \cdot 2 \cdot 1 (n-4)!}$$



$$\frac{n(n-1)(n-2)(n-3)}{24}$$

$${}^nC_{10} = \frac{12 \times 11}{2!}$$

$${}^nC_{10} = \frac{12 \cdot 11}{2!} = \frac{12 \cdot 11 \cdot 10!}{2! \times 10!} = \frac{12!}{2!(12-2)}$$

$${}^nC_{10} = {}^{12}C_2$$

$${}^nC_{n-10} = {}^{12}C_2$$

$$n-10=2 \Rightarrow n=12$$

$${}^5C_2$$

$$\frac{5!}{2!} = \frac{5 \cdot 4 \cdot 3!}{2!} = \frac{5 \cdot 2}{1} = 10$$

$$\frac{5!(5-2)!}{2! \cdot 3!}$$

$${}^6C_3$$

$$\frac{6!}{3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3!} = 2 \cdot 5 \cdot 2 = 20$$

$$\frac{6!(6-3)!}{3! \cdot 3!} = \frac{6!}{3! \cdot 3!}$$

$${}^{16}C_{11}$$

$$\frac{16!}{11!} = \frac{16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11!}{11!} = 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 = 36960$$

$$\frac{16!(16-11)!}{11! \cdot 5!} = \frac{16!}{11! \cdot 5!}$$

$${}^{17}C_{11}$$

$$\frac{17!}{11!} = \frac{17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11!}{11!} = 17 \cdot 16 \cdot 15 \cdot 14 \cdot 13 \cdot 12 = 12376320$$

$$\frac{17!(17-11)!}{11! \cdot 6!} = \frac{17!}{11! \cdot 6!}$$

How many (a) diagonals and (b) triangles can be formed by joining the vertices of the polygon having.

- ① 5 sides ② 8 sides ③ 12 sides

(a) Diagonals

Number of diagonals of 5 sided polygon:

$${}^5C_2 - 5 = \frac{5!}{2!} - 5 = \frac{5 \cdot 4 \cdot 3!}{2! \cdot 2!} - 5$$

$$= \frac{5 \cdot 4}{2 \cdot 1} - 5 = \frac{20}{2} - 5 = 10 - 5 = 5$$

Number of 8 sided polygon

$$= {}^8C_2 - 8 = \frac{8!}{2!} - 8 = \frac{8 \cdot 7 \cdot 6!}{2! \cdot 6!} - 8$$

$$= \frac{8 \cdot 7}{2} - 8 = 28 - 8 = 20$$

Number 12 sided polygon

$$= {}^{12}C_2 - 12 = \frac{12!}{2!} - 12 = \frac{12 \cdot 11 \cdot 10!}{2! \cdot 10!} - 12$$

$$= \frac{12 \cdot 11}{2} - 12 = 66 - 12 = 54$$

$$66 - 12 = 54$$



(b) triangles

$${}^5C_3$$

$$= \frac{5!}{(5-3)! 3!} = \frac{5 \cdot 4 \cdot 3!}{2! 3!} = \frac{10}{2} = 10$$

$${}^8C_3$$

$$= \frac{8!}{(8-3)! 3!} = \frac{8 \cdot 7 \cdot 6!}{5! 3!} = \frac{8 \cdot 7 \cdot 2!}{2! 3!} = \frac{28}{1} = 28$$

$${}^{12}C_3$$

$$= \frac{12!}{(12-3)! 3!} = \frac{12 \cdot 11 \cdot 10 \cdot 9!}{9! 3!} = \frac{12 \cdot 11 \cdot 10}{2} = 66$$

The members of a club are 12 boys and 8 girls. In how many ways can a committee of 3 boys and 2 girls be formed?

$${}^{12}C_3 \text{ ways} \times {}^8C_2 \text{ ways}$$

$$= \frac{12!}{3! (12-3)!} \times \frac{8!}{2! (8-2)!}$$

$$= \frac{12 \cdot 11 \cdot 10 \cdot 9!}{3! 9!} \times \frac{8 \cdot 7 \cdot 6!}{2! 6!}$$

$$= 220 \times 28 = 6160$$

$$\begin{aligned}
 &= 12 \cdot 11 \cdot 10 \times 8 \cdot 7 \\
 &= 3 \cdot 2 \cdot 1 \times 2 \\
 &= 498 \times 56 \\
 &= 220 \times 28 = 6160
 \end{aligned}$$

How many committee of 5 members can be chosen from a group of 8 person when each committee must include 2 particular persons? We have select 3 members out of 6 person and this can be done in  ${}^6C_3$  ways.

$$\frac{6!}{(6-3)!3!} = \frac{6 \cdot 5 \cdot 4 \cdot 3!}{3! \cdot 3 \cdot 2 \cdot 1} = 5 \cdot 4 = 20$$

## Circular permutation

Circular permutation refers to arrangement of object in a circle where the position of object are considered equivalent if one can be rotated into another.

$$P = (n-1)!$$

$$P = (n-1)!$$



Exercise:-

Find the number of different ways that a family of 6 can be seated around a circular table with 6 chairs.

$$\begin{aligned}P &= (n-1)! \\&= (6-1)! \\&= 5! \\&= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\&= 120 \text{ ways.}\end{aligned}$$

In how many ways can you arrange 5 different colored beads in a bracelets?

$$P = (n-1)!$$

$$= \frac{5!}{2}$$

$$= \frac{4!}{2}$$

$$= 4 \cdot 3 \cdot 1 = 12 \text{ ways}$$

$$= \frac{4 \cdot 3 \cdot 2 \cdot 1}{2}$$

There are 8 people in a dinner gathering. In how many ways can the host (one of the 8) arrange his guests around a dining table if

- (a) they can sit on any of the chairs?  
 (b) 3 people insist on sitting beside each other.  
 (c) 2 people refuse to sit beside each other.

$$n = 8$$

$$\begin{aligned} (a) \quad P &= (n-1)! \\ &= (8-1)! \\ &= 7! \\ &= 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 5040 \end{aligned}$$

$$5+1 = 6$$

$$\begin{aligned} (b) \quad P &= (6-1)! \times 3! \\ &= 5! \times 3! \\ &= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 3 \cdot 2 \cdot 1 \\ &= 720 \end{aligned}$$

$$\begin{aligned} (c) \quad &= (7-1)! \cdot 2! \\ &= 6! \cdot 2! \\ &= 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1 \\ &= 1440 \end{aligned}$$



How many ways can 7 people be seated at a round table if 3 people refuse to sit next to each other.

$$P = (n-1)$$

$$P = (7-1)! = 6!$$

Now, let us assume these three particular people always sit together

A B C D (EFG)

$$(5-1)! \cdot 3! \cdot 3$$

$$(4)! \cdot 3!$$

So therefore,

$$= 6! - (4)! \cdot (3)!$$

$$= 720 - 144$$

$$= 576 \text{ ways}$$

How many ways can 4 people sit around a circular table?

$$n = 4$$

$$(n-1)! = (4-1)!$$

$$= 3 \cdot 2 \cdot 1 = 6$$

How many way can 4 boys 2 girls be seated at a round table?

$$n = 6$$

$$P = (n-1)$$

$$P = (6-1)$$

$$= 5!$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 120$$

How many way can 3 girls and 5 boys be seated around a circular table if all girls must sit together?

$$3 \text{ girls} = 1 \text{ group}$$

$$1 \text{ group seat and 5 seats}$$

$$n = 6$$

$$= (n-1)! = (6-1)!$$

$$3 \text{ girls} = 3!$$

$$= (6-1)! \cdot 3!$$

$$= 5! \cdot 3!$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 3 \cdot 2 \cdot 1$$

$$= 720$$



How many ways can 5 girls and 5 boys be seated at a round table if No restriction

$$P = (n-1)!$$

$$P = (10-1)!$$

$$= 9!$$

$$= 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 362,880 \text{ ways}$$

A circular table has 5 chairs  
How many different seating arrangements are possible for 5 people?

$$P = (n-1)!$$

$$P = (5-1)!$$

$$= 4!$$

$$= 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 24$$

A music band has 6 members  
They want to stand in a circular formation on stage. How many different arrangements are possible?

$$P = (n-1)!$$

$$(6-1)!$$

$$= 5!$$

$$= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 120$$

A company has 8 employees. They want to arrange themselves in a circular formation for a team photo. How many different arrangements are possible?

$$P = (n-1)!$$

$$= (8-1)!$$

$$= 7!$$

$$= 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 5040$$

A circular track has 10 lanes. How many different arrangements are possible for 10 athletes running in a circular formation?

$$P = (n-1)!$$

$$= (10-1)!$$

$$= 9!$$

$$= 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 362880$$



A circular table has 9 chairs. How many different seating arrangements are possible for 9 people if 3 of the people ~~for 9~~ are family and want to sit together?

$$\frac{(9-1)}{(3-1)} = \frac{8!}{2!} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1}$$

$$= 20160$$

A group of 7 friends want to sit around a circular table. However, 2 of the friends are rivals and cannot sit next to each other. How many different seating arrangements are possible?

$$= (7-1)! - (2 \times 5!)$$

$$= 6! - 240$$

$$= 360$$

A circular track has 12 lanes  
How many different arrangements  
are possible for 12 athletes running  
in a circular formation if 4 of  
the athletes are from the same  
team and want to run  
together?

$$= \frac{(12-1)!}{(4-1)!}$$

$$= \frac{11!}{3!}$$

$$= \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1}$$

$$= 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4$$

$$= 1320$$